Computer Network
The Medium Access Sub layer

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Random Access Control

- **ALOHA(Pure and Slotted ALOHA)**
- **Carrier Sense Multiple Access**
- **Carrier Sense Multiple Access with Collision Detection**
- **Carrier Sense Multiple Access with Collision Avoidance**

Controlled Access Reservation, Polling and Token Passing

Channelization FDMA, TDMA , CDMA

Data Link Layer Structure

The Data Link Layer is responsible for transmission of data between two nodes. Its main functions are-

- Data Link Control
- Multiple Access Control

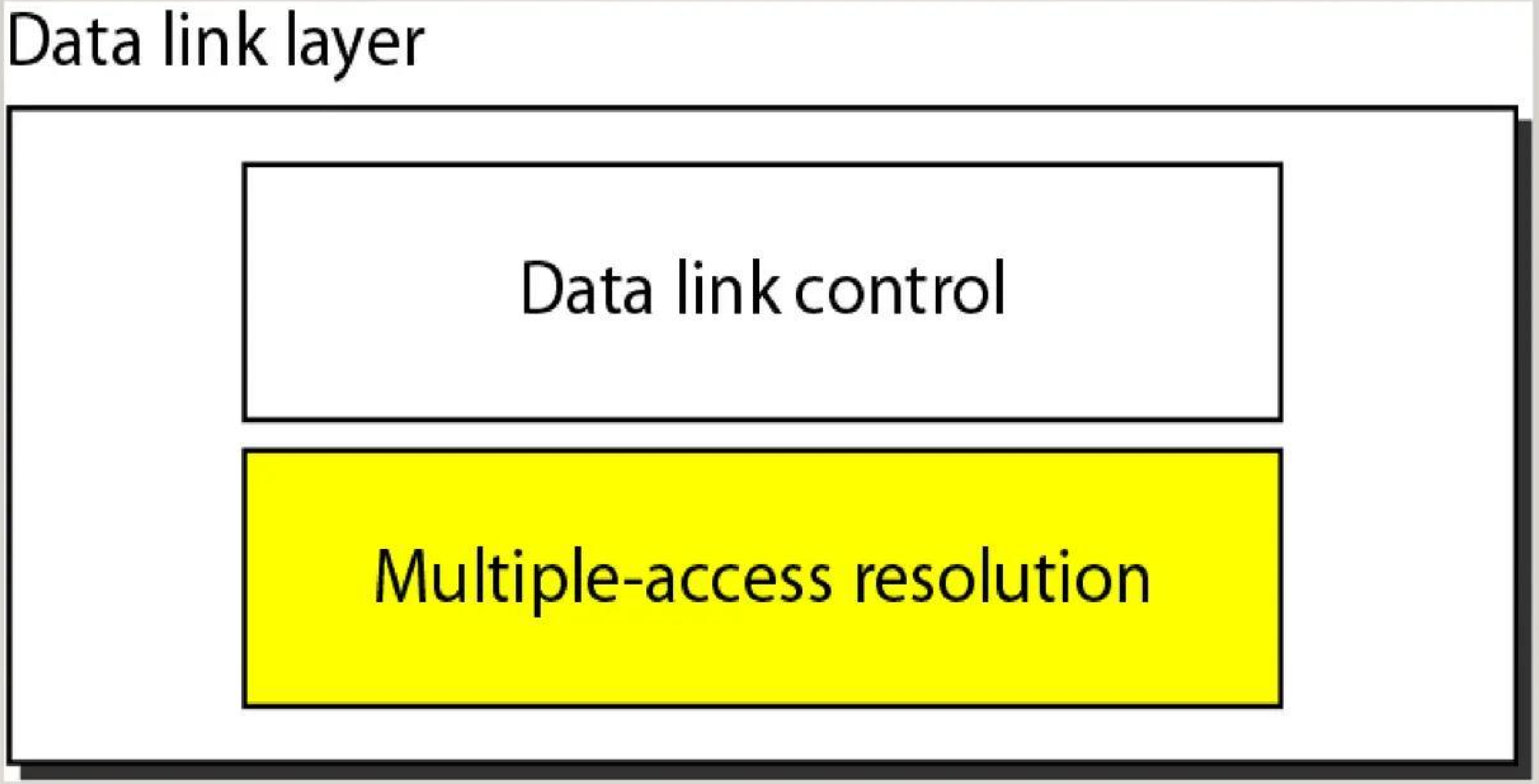


Figure 1 Data Link Layer Strcuture

Types of Multiple access protocols

Multiple Access Protocols are a set of rules used in computer networks to control how multiple devices share and access a common communication channel.

Following are the types of Multiple Access Protocols, which are separated into many different processes, are listed below

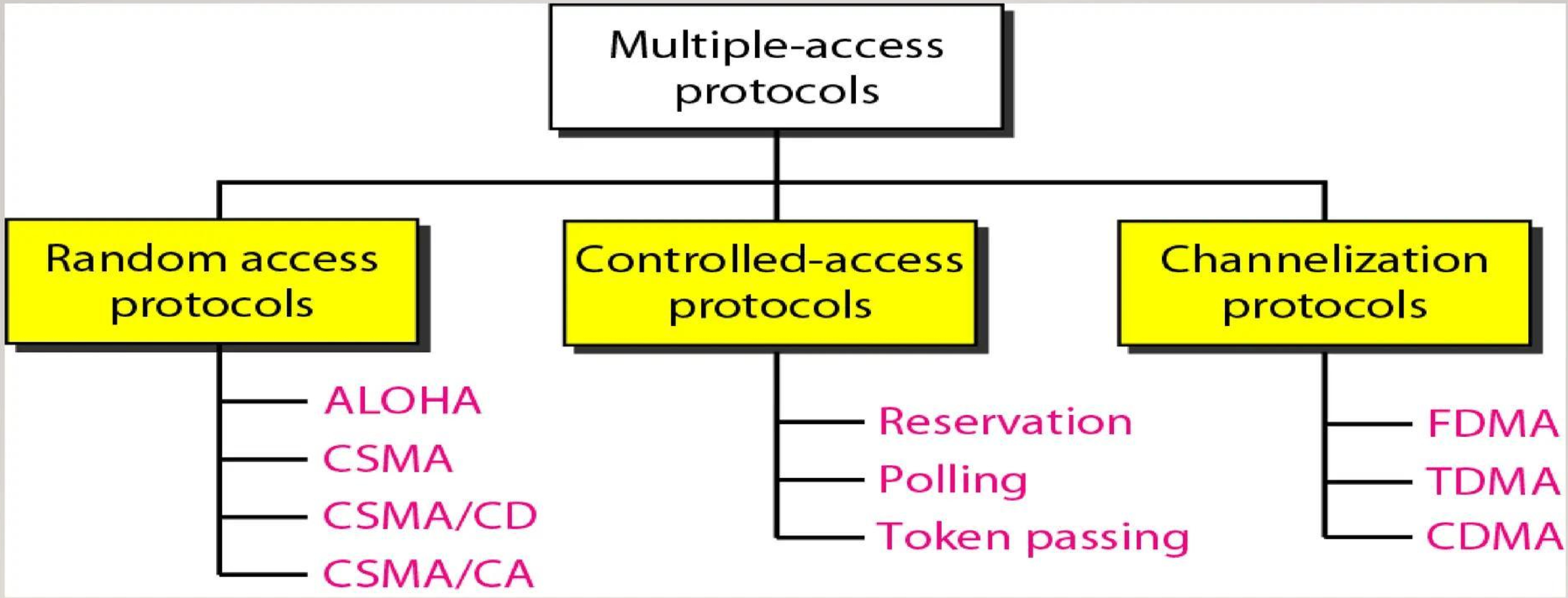


Figure 2 Types for Multiple Access Protocols

Random Access

In Random access or Contention methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.

Types of Random Access

- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

pure ALOHA

- Pure ALOHA uses a simple idea that is to let user transmit whenever they have to send.
- Pure ALOHA having feature with the feedback property that enables the channel and finds the whether frame is destroyed.
- Feedback in LANs immediate.
- Pure ALOHA requires acknowledgement.
- Pure ALOHA provides 18 percent of channel utilization.

Frames in a pure ALOHA network

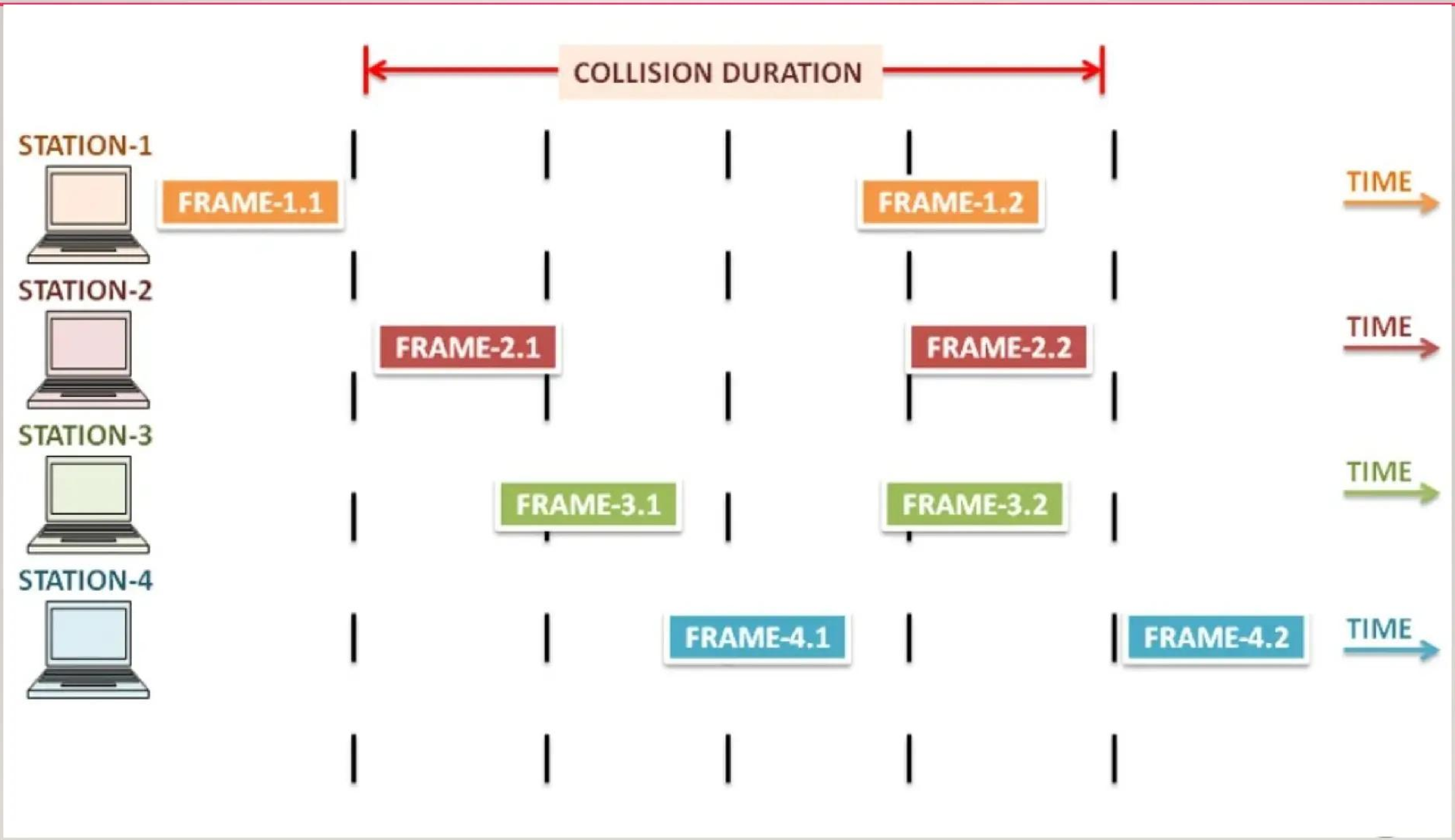


Figure 3 Procedure for pure ALOHA protocol

Procedure for pure ALOHA protocol

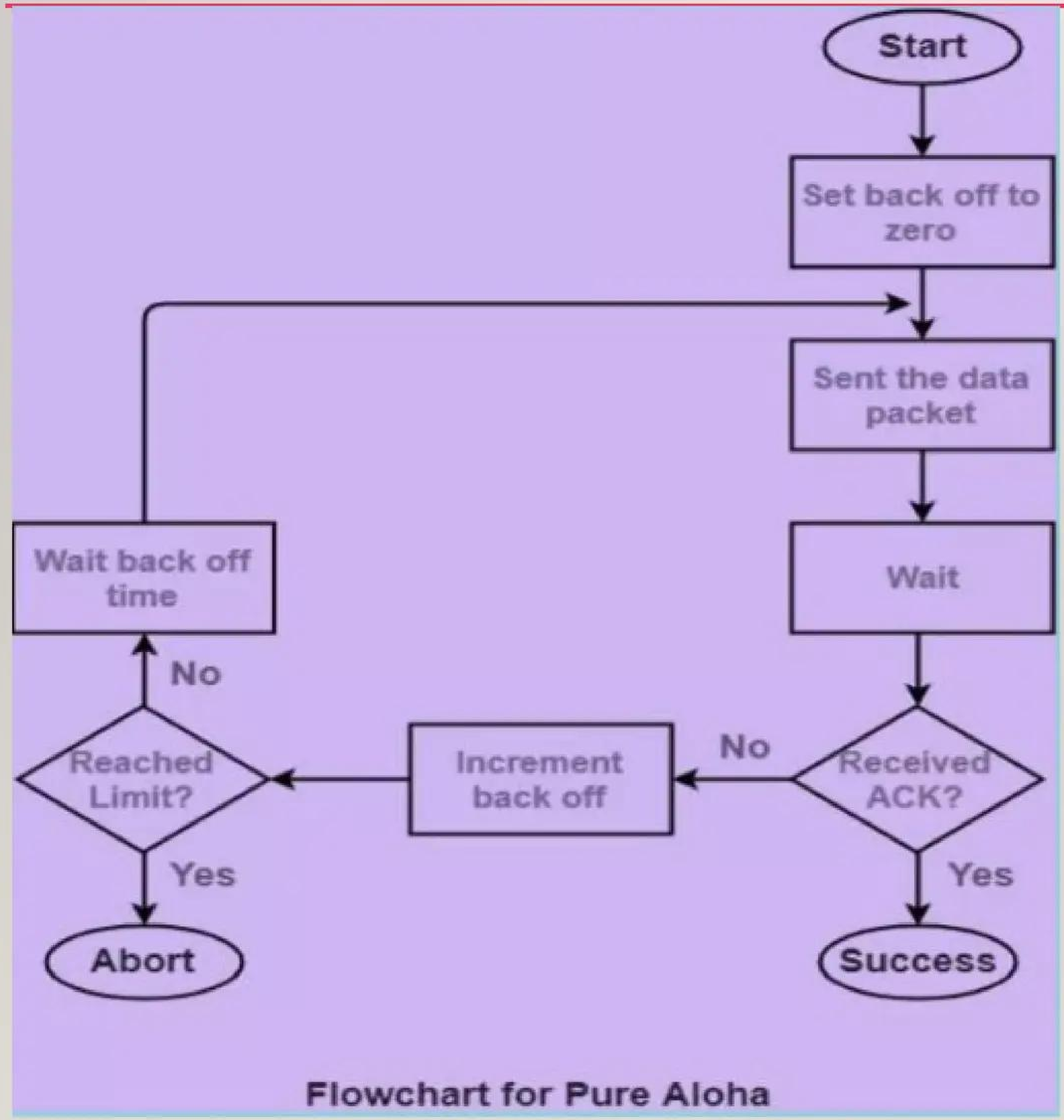


Figure 4 Flow chart for pure Aloha

Slotted ALOHA network

As the name suggests, in the slotted ALOHA the time of the shared channel is simply divided into discrete intervals that are commonly known as Time Slots.

- In Slotted ALOHA it is imposed on each station to send the data only at the beginning of the time slot.
- As a station is allowed to send the data only at the beginning of the time slot, in case if any station misses this moment then it must have to wait until the beginning of the next time slot.
- In case if two stations try to send at the beginning of the same time slot then there are chances for the occurrence of the collision

Figure 6 Frames in a slotted ALOHA network

Frames in a slotted ALOHA network

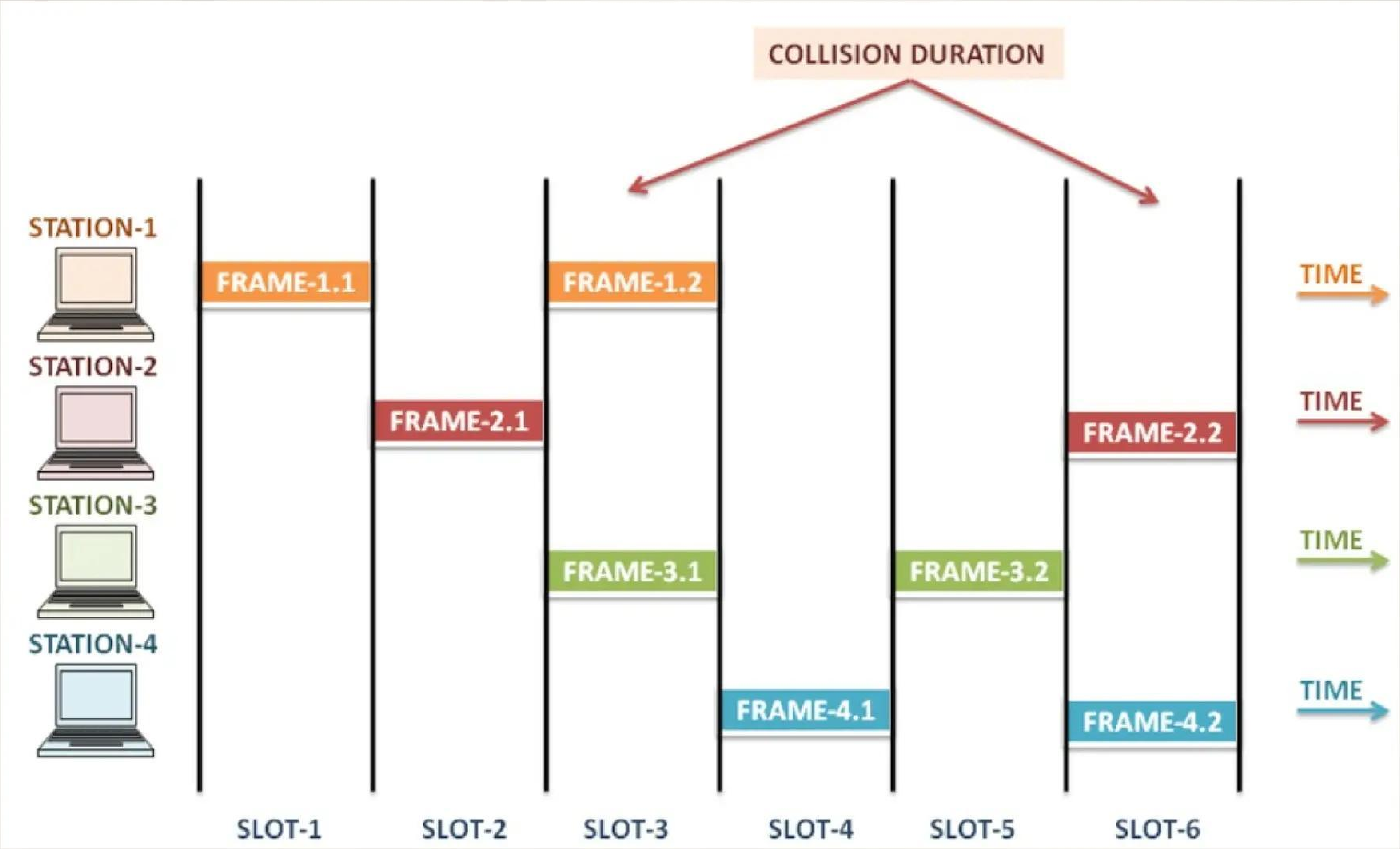


Figure 5 Frames in a slotted ALOHA network

Pure and slotted ALOHA network

Feature	Pure Aloha	Slotted Aloha
Transmission Time	Anytime (Continuous)	Beginning of slots
Time Structure	Continuous	Discrete (Slots)
Synchronization	Not Required	Required
Max Efficiency	18.4%	36.8%
Collisions	Higher	Lower (Half of Pure)
Complexity	Low	Higher

CSMA

Fundamental Concepts of CSMA

The primary goal of CSMA is to avoid collisions, which occur when two or more devices transmit data simultaneously over the same channel. Key concepts in CSMA include:

Carrier Sense: Before transmitting, a device checks the carrier signal on the channel to determine if it is free (idle) or busy. This is the "sense" part of CSMA.

Multiple Access: Multiple devices have access to the same communication channel. CSMA helps coordinate this access to avoid collisions.

Collision Detection and Avoidance: CSMA includes mechanisms to detect collisions when they occur and take steps to minimize their likelihood.

Types of CSMA

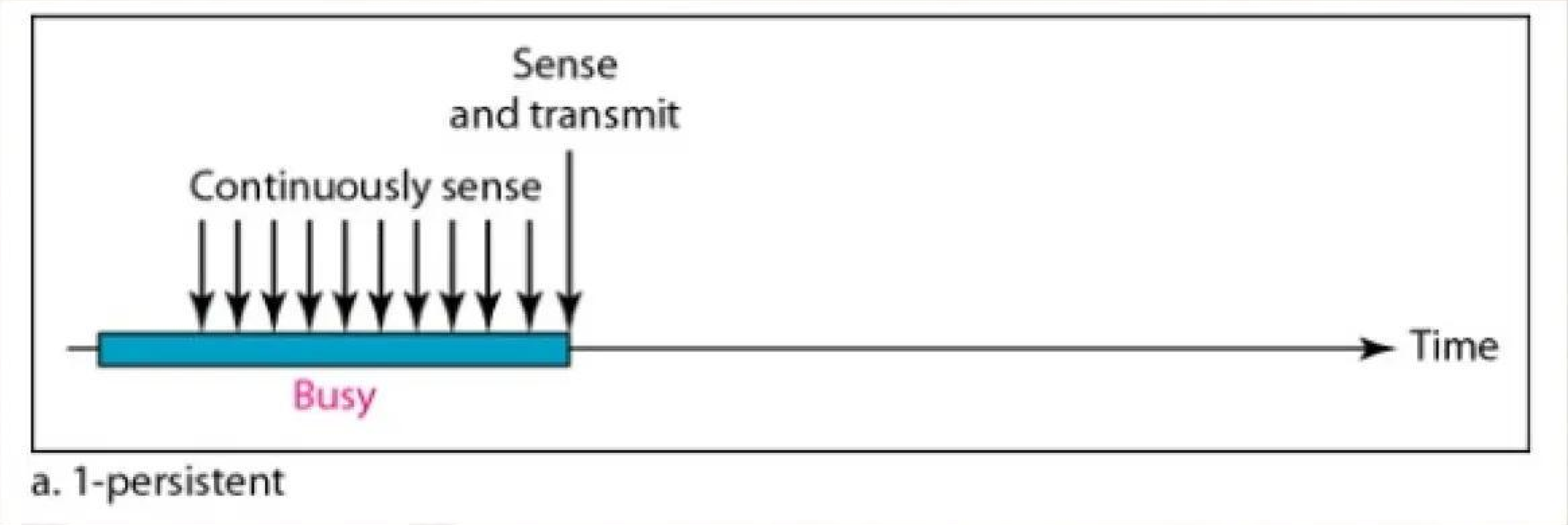
There are several variations of CSMA, each with different mechanisms for handling potential collisions:

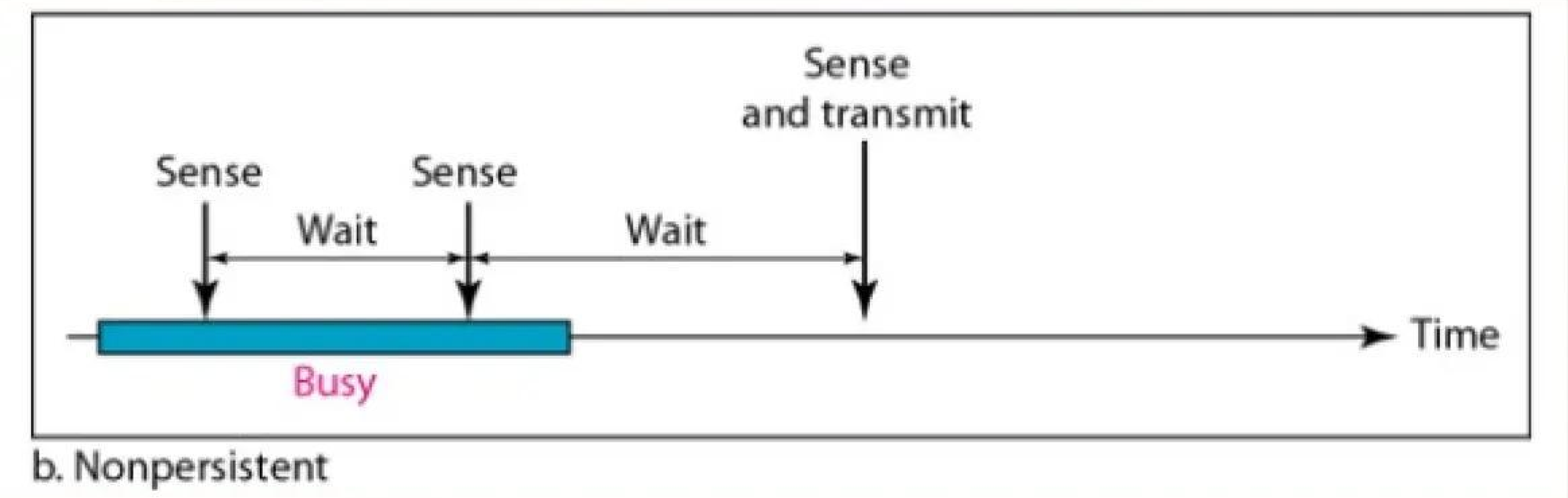
1-persistent CSMA: In this method, a device continuously senses the channel and transmits immediately when the channel is found to be free. If the channel is busy, the device waits until it becomes free and then transmits. This approach can lead to higher collision rates, especially in highly congested networks.

Non-persistent CSMA: Here, if the channel is busy, the device waits for a random amount of time before sensing the channel again. This reduces the chances of collisions compared to 1-persistent CSMA, but can result in longer delays.

P-persistent CSMA: This is a hybrid approach used in slotted channels (time is divided into discrete intervals). When the channel is free, the device transmits with a probability ‘p’ or waits for the next slot with a probability ‘1-p’. This method balances the need for immediate transmission and collision avoidance.

Behavior of three CSMA persistent methods





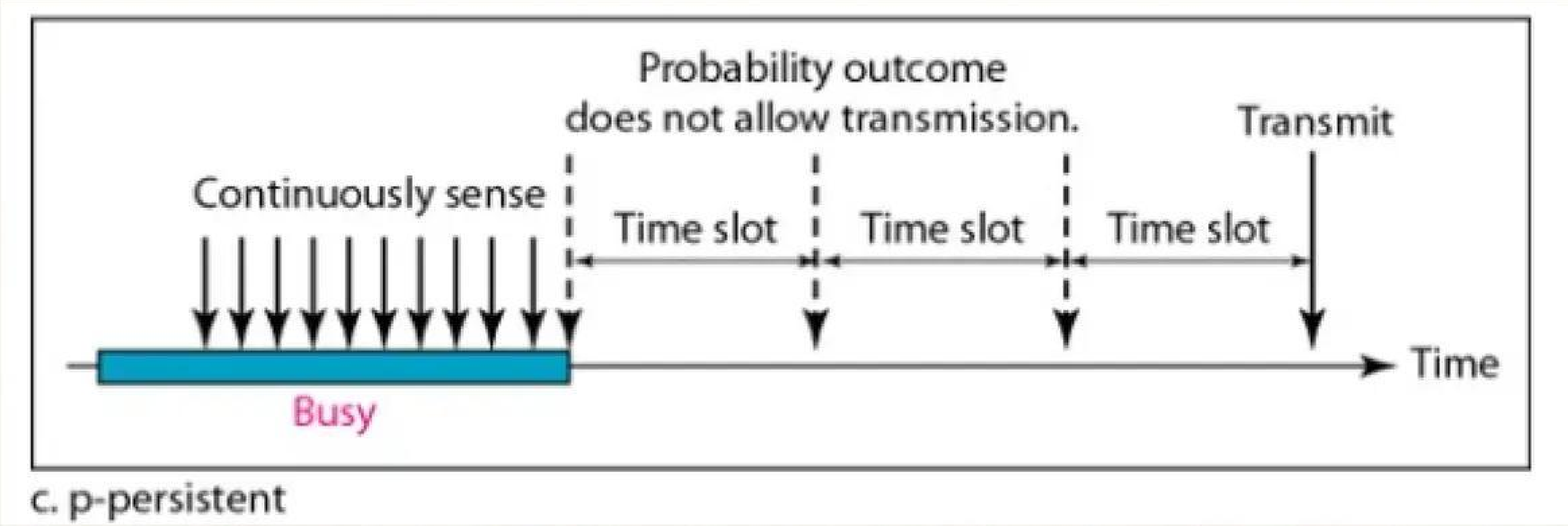


Figure 6 CSMA persistent methods

Flow diagram of three CSMA persistent methods

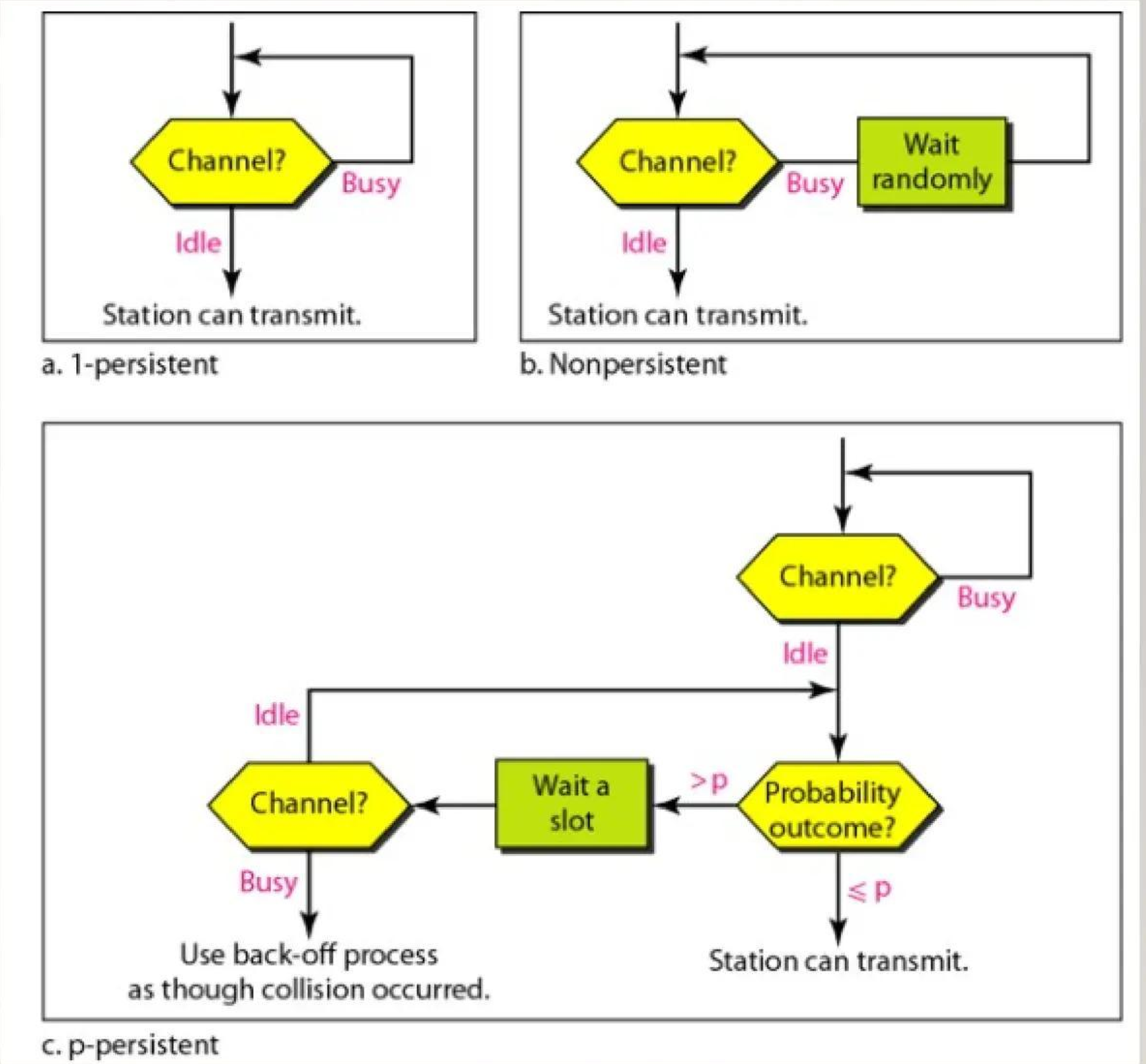


Figure 7 Flow chart of CSMA persistent methods

Space/time model of the collision in CSMA

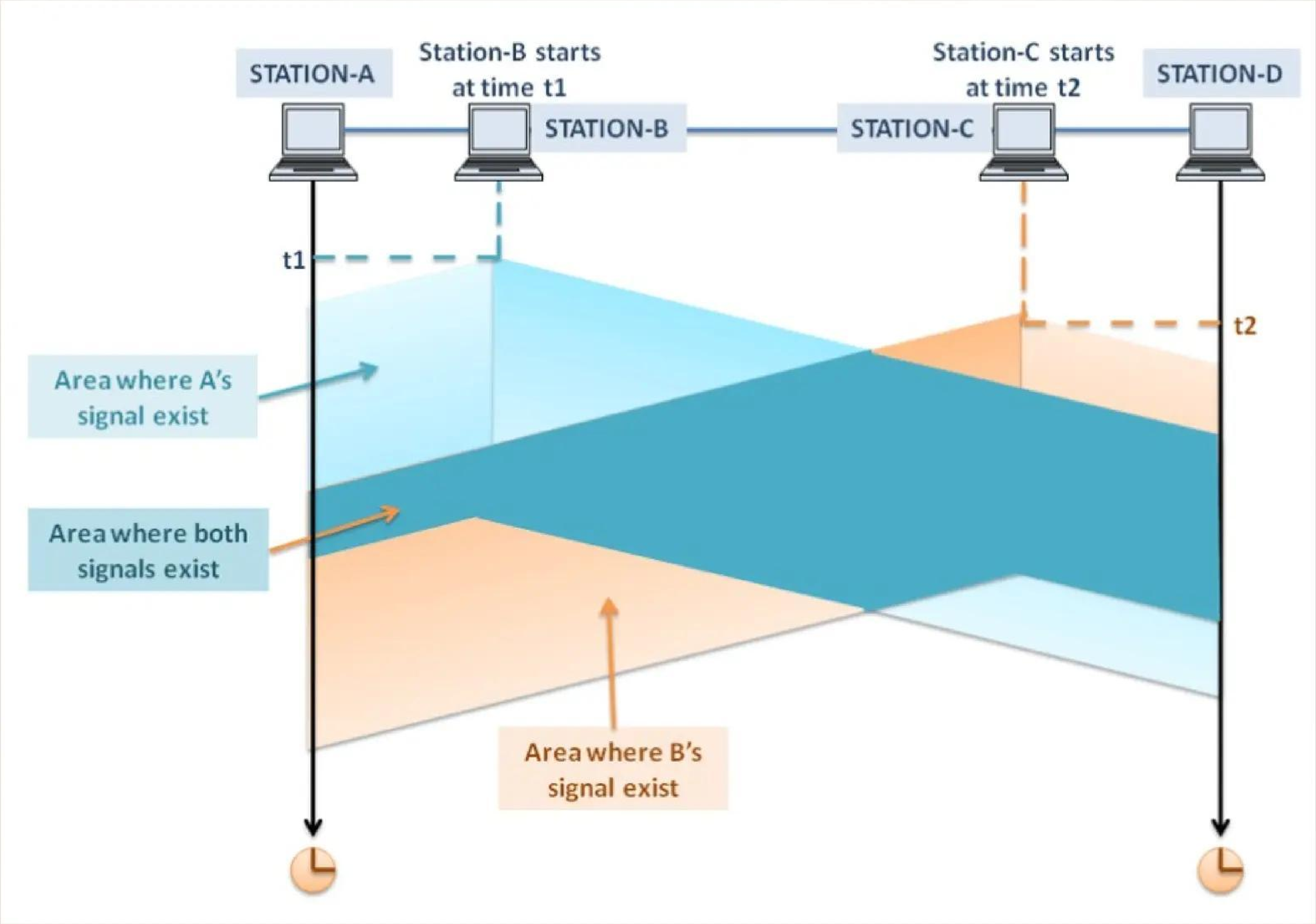


Figure 8 Space/time model of the collision in CSMA

CSMA/CD (Collision Detection)

CSMA/CD is an extension of CSMA used primarily in wired networks like Ethernet. It includes mechanisms for detecting collisions and taking corrective actions. The key steps in CSMA/CD are:

- **Carrier Sensing:** Devices monitor the channel to check if it is free before transmitting.
- **Transmission:** If the channel is free, the device starts transmitting data.
- **Collision Detection:** Devices continue to monitor the channel while transmitting. If a collision is detected (by sensing interference or a drop in signal quality), the device immediately stops transmitting.
- **Jamming Signal:** The device sends a jamming signal to inform other devices of the collision.
- **Backoff Algorithm:** After a collision, devices wait for a random period before attempting to retransmit. The waiting time typically increases exponentially with each subsequent collision.

Collision of the first bit in CSMA/CD

Figure 9 Collision of the first bit in CSMA/CD

Collision and abortion in CSMA/CD

Figure 10 Collision and abortion in CSMA/CD

Flow diagram for the CSMA/CD

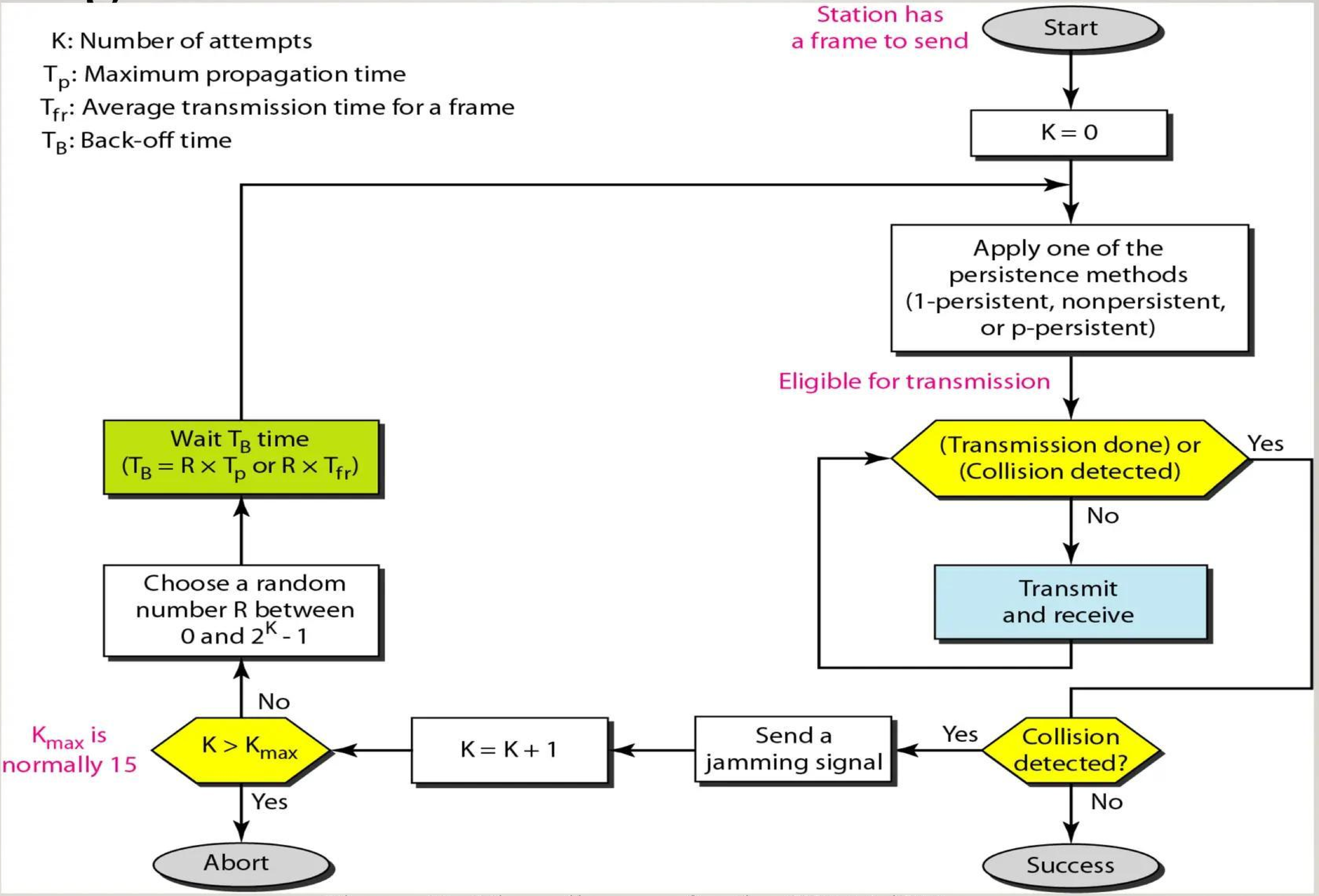


Figure 11 Flow diagram for the CSMA/CD

CSMA/CD

Advantages of CSMA CD

Some of the advantages of CSMA/CD are:

- It is simple and inexpensive to implement, as it does not require complex hardware or software.
- It is efficient and fair, as it allows every device to have an equal chance of transmitting data when the medium is free.
- It is robust and adaptable, as it can handle variable traffic and network conditions and recover from collisions quickly.

Disadvantages of CSMA CD

- The efficiency of CSMA CD decreases with distance, making it inappropriate for long-distance networks.
- The performance of collision detection is negatively impacted when a large number of devices are introduced to a Carrier Sense Multiple Access with Collision Detection (CSMA CD) system.

CSMA/CA (Collision Avoidance)

CSMA/CA is predominantly used in wireless networks, such as Wi-Fi, where collision detection is more challenging due to the nature of wireless communication. Instead of detecting collisions, CSMA/CA focuses on avoiding them. The key steps are:

- **Carrier Sensing:** Devices listen to the channel before transmitting.
- **Collision Avoidance:** If the channel is busy, the device waits for a random backoff period before attempting to sense the channel again. Some implementations use a Request to Send (RTS) and Clear to Send (CTS) mechanism to reserve the channel for a specific communication.
- **Transmission:** If the channel is free, the device transmits the data.

Timing in CSMA/CA

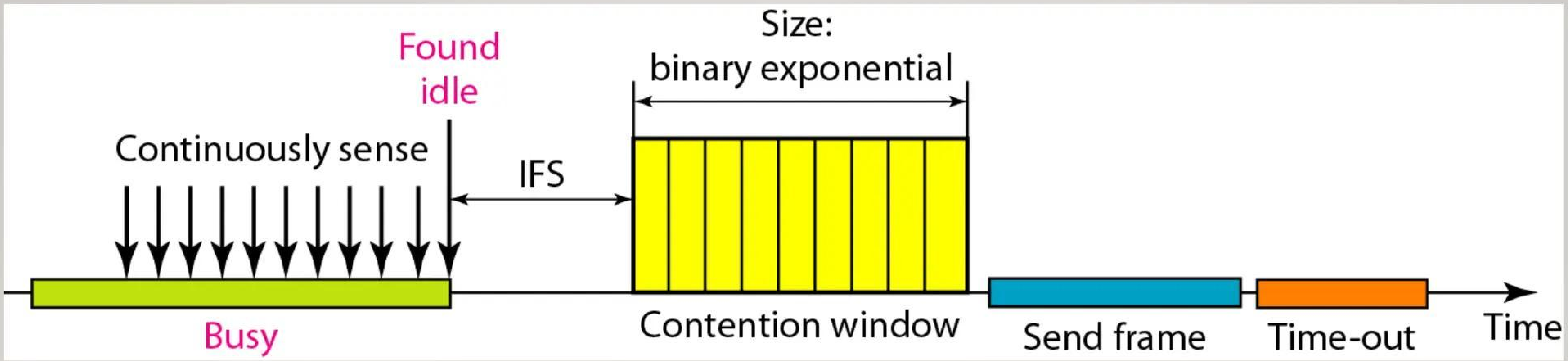


Figure 13 Timing in CSMA/CA

IFS in CSMA/CA

In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Congestion in CSMA/CA

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window;
it stops the timer and restarts it when the channel becomes idle.

Flow diagram for CSMA/CA

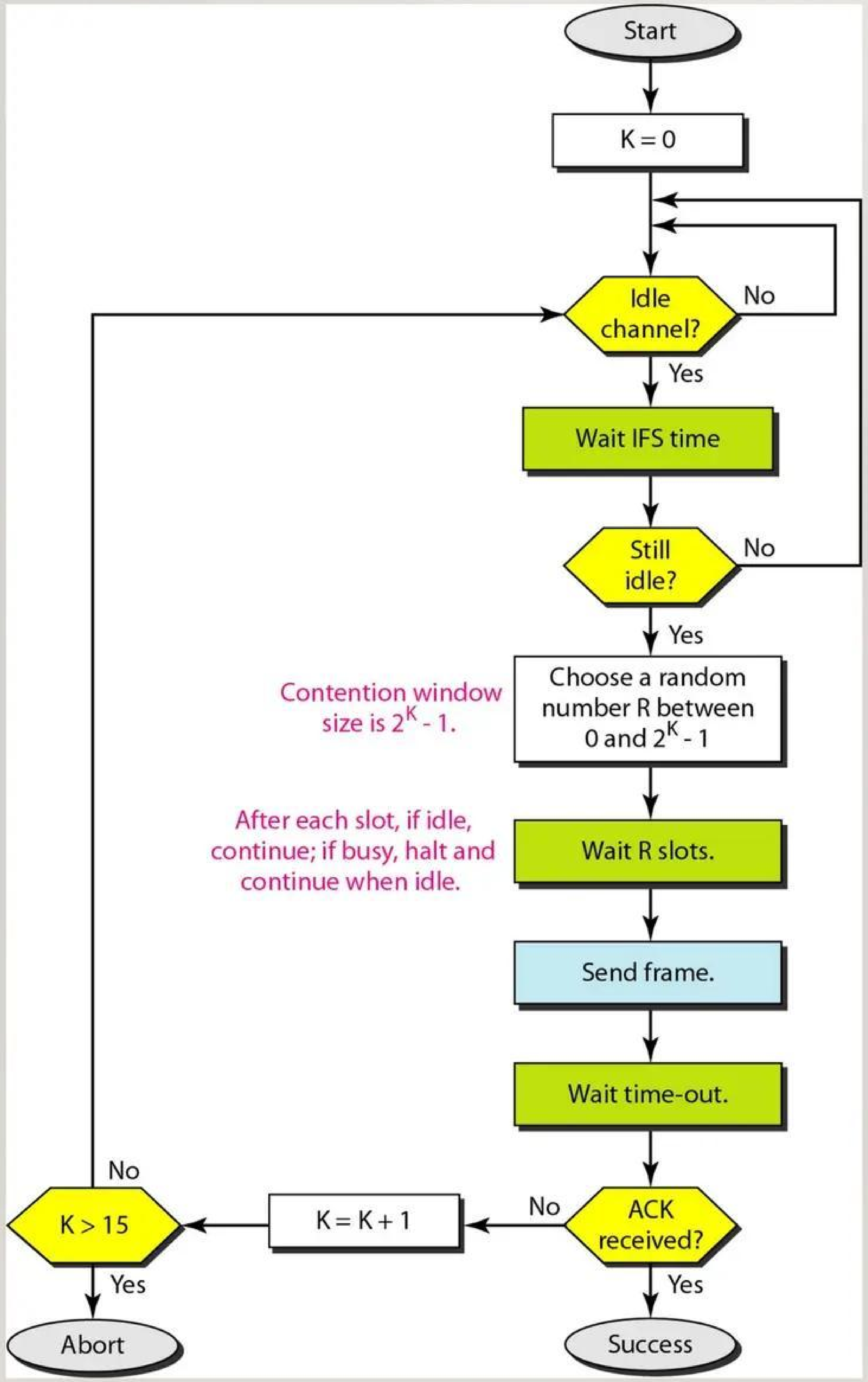


Figure 14 Flow diagram for CSMA/CA

CSMA/CA

Advantages of CSMA CA

Some of the advantages of CSMA/CA are:

- When data packets are large in size, collisions are less likely to occur in CSMA CA.
- It takes care of the data packets and sends them off to their destination as necessary.
- Instead of detecting collisions on the shared channel, it is used to avoid them.
- CSMA CA ensures that no transmitted data is ever lost or wasted.
- The RTS/CTS extensions are used to cut down on unnecessary network traffic.

Disadvantages of CSMA CA

- Compared to CSMA CD, it is less effective.
- Each station utilizes a larger amount of bandwidth.
- On some occasions, the CSMA/CA protocol shows a delay that is double the duration of the average amount of time required to transmit a data packet.

CONTROLLED ACCESS PROTOCOLS

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.

- Reservation
- Polling
- Token Passing

RESERVATION ACCESS PROTOCOL

In reservation-based protocols, devices must reserve the right to transmit data before actually sending it. This reservation process ensures that only one device has access to the channel at a time.

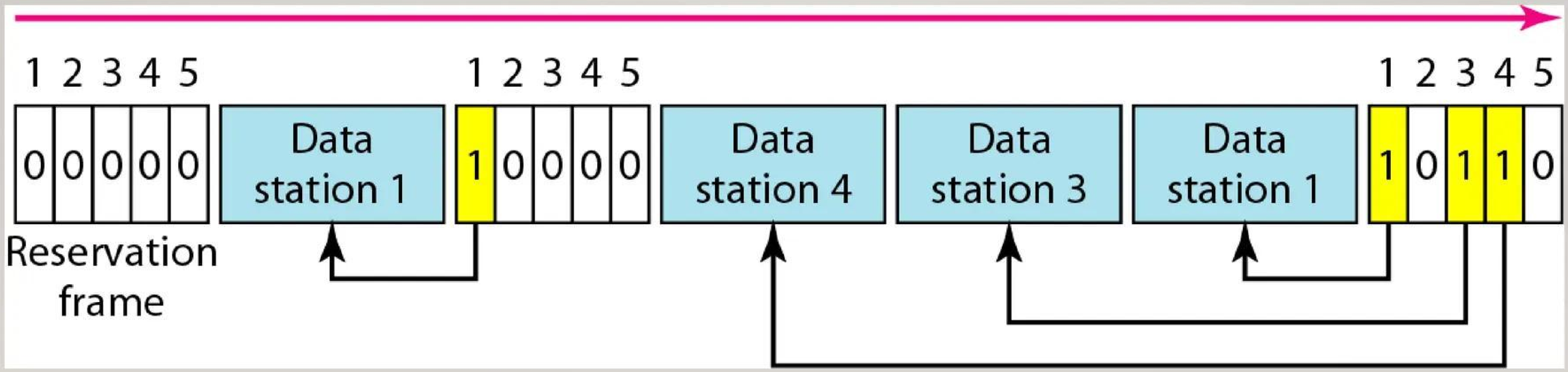


Figure 15 Reservation access method

POLLING ACCESS PROTOCOL

In a polling system, a central controller polls each device in turn, giving them an opportunity to transmit. If the device has data to send, it does so; otherwise, the controller moves on to the next device.

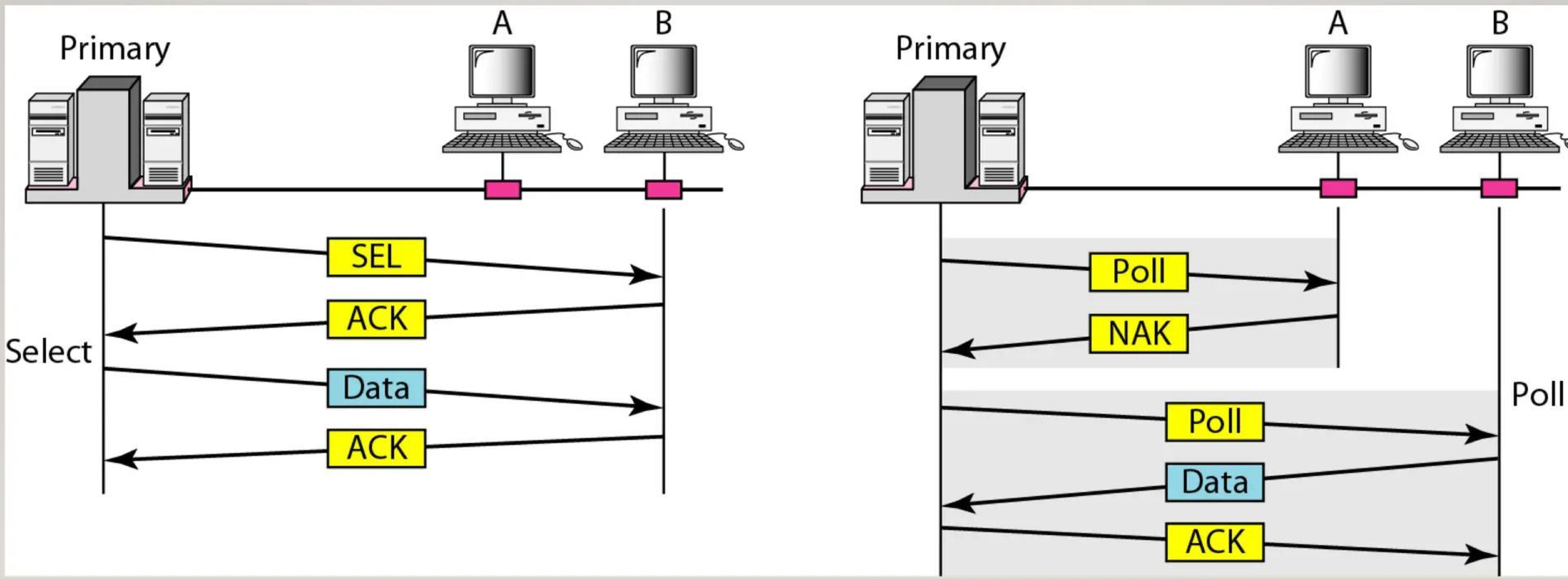


Figure 16 Select and poll functions in polling access method

TOKEN PASSING PROTOCOL

Token passing is a method where a special data packet, called a "token," circulates among the devices. Only the device holding the token is allowed to transmit data. After transmission, the token is passed to the next device.

TOKEN PASSING PROTOCOL

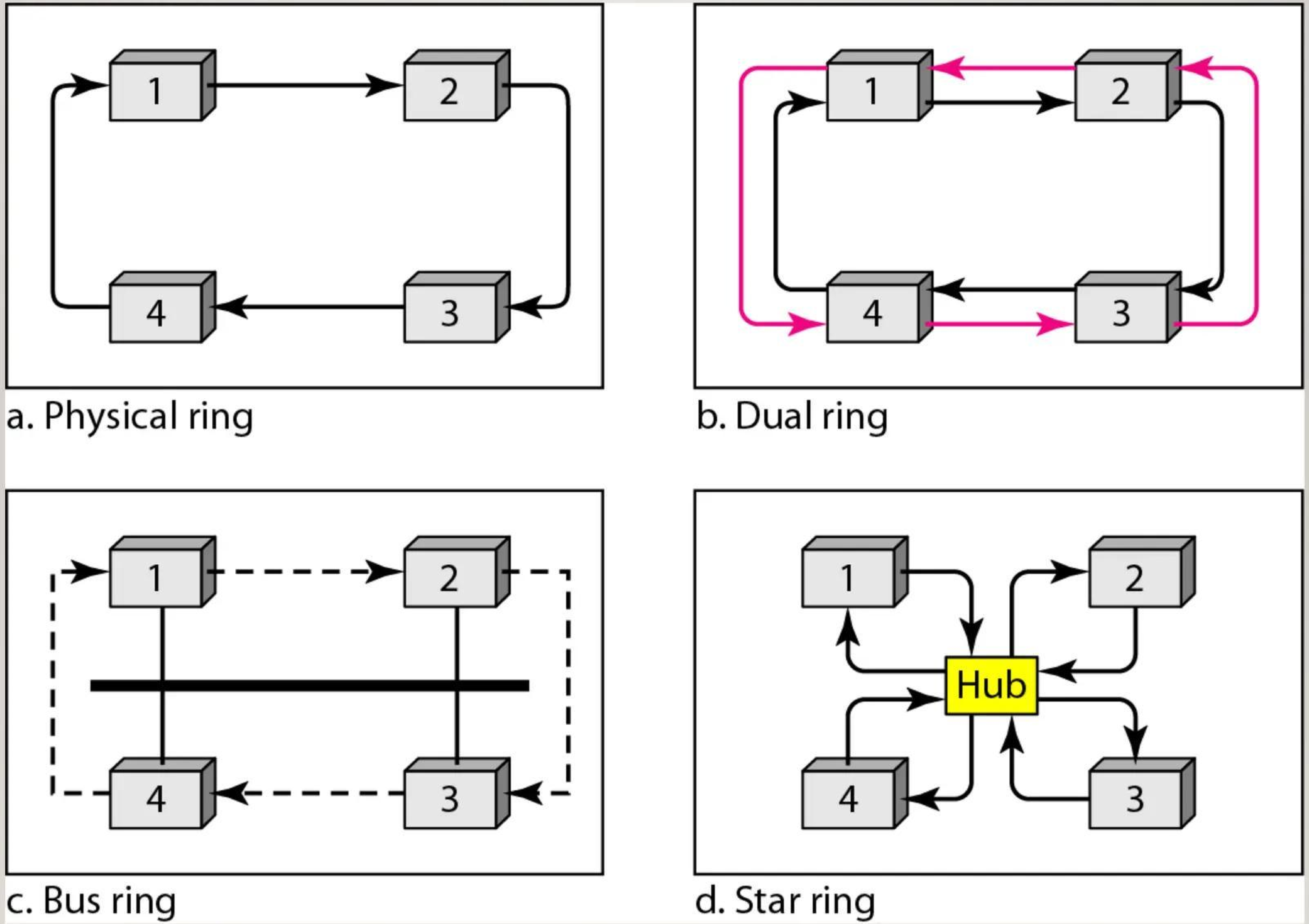


Figure 17 Logical ring and physical topology in token-passing access method

CHANNELIZATION

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols.

- FDMA
- TDMA
- CDMA

Frequency-division multiple access (FDMA)

- Total system bandwidth is divided into narrow frequency slots. Each user is allocated a unique frequency band or channel
- A user is free to transmit or receive all the time on its allocated radio channel, but the cost of transceiver is high, as each has to be designed on a different band

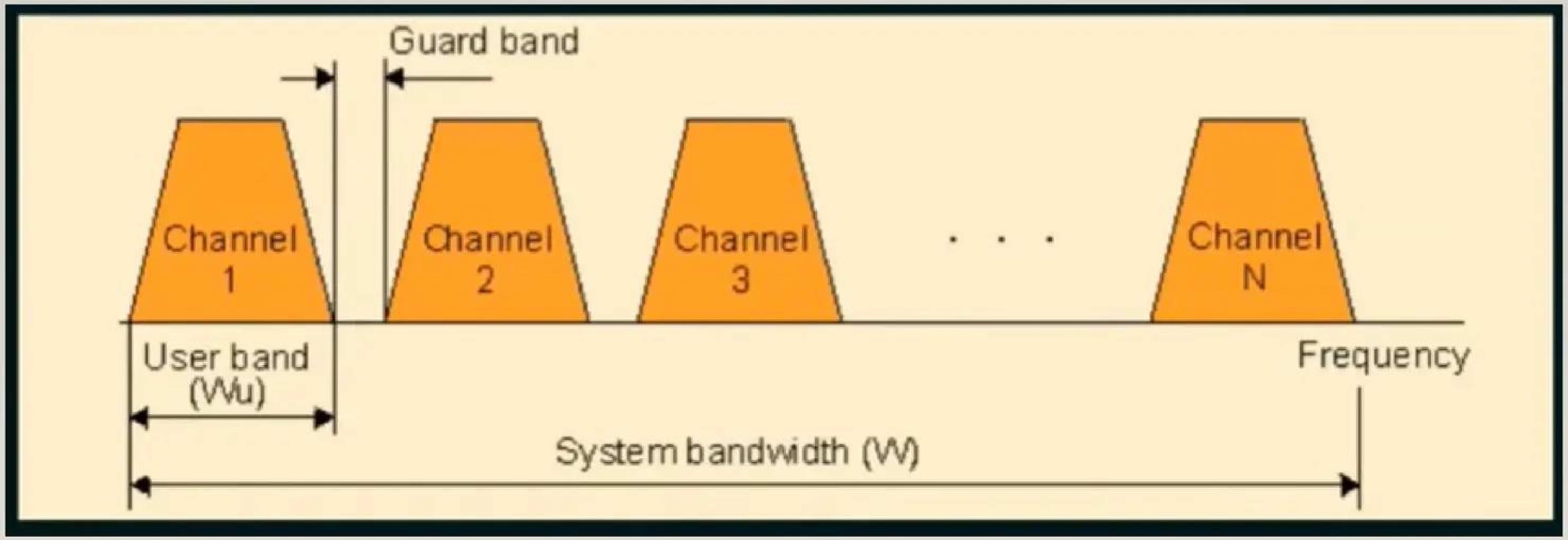


Figure 18 Frequency-division multiple access (FDMA)

Frequency-division multiple access (FDMA)

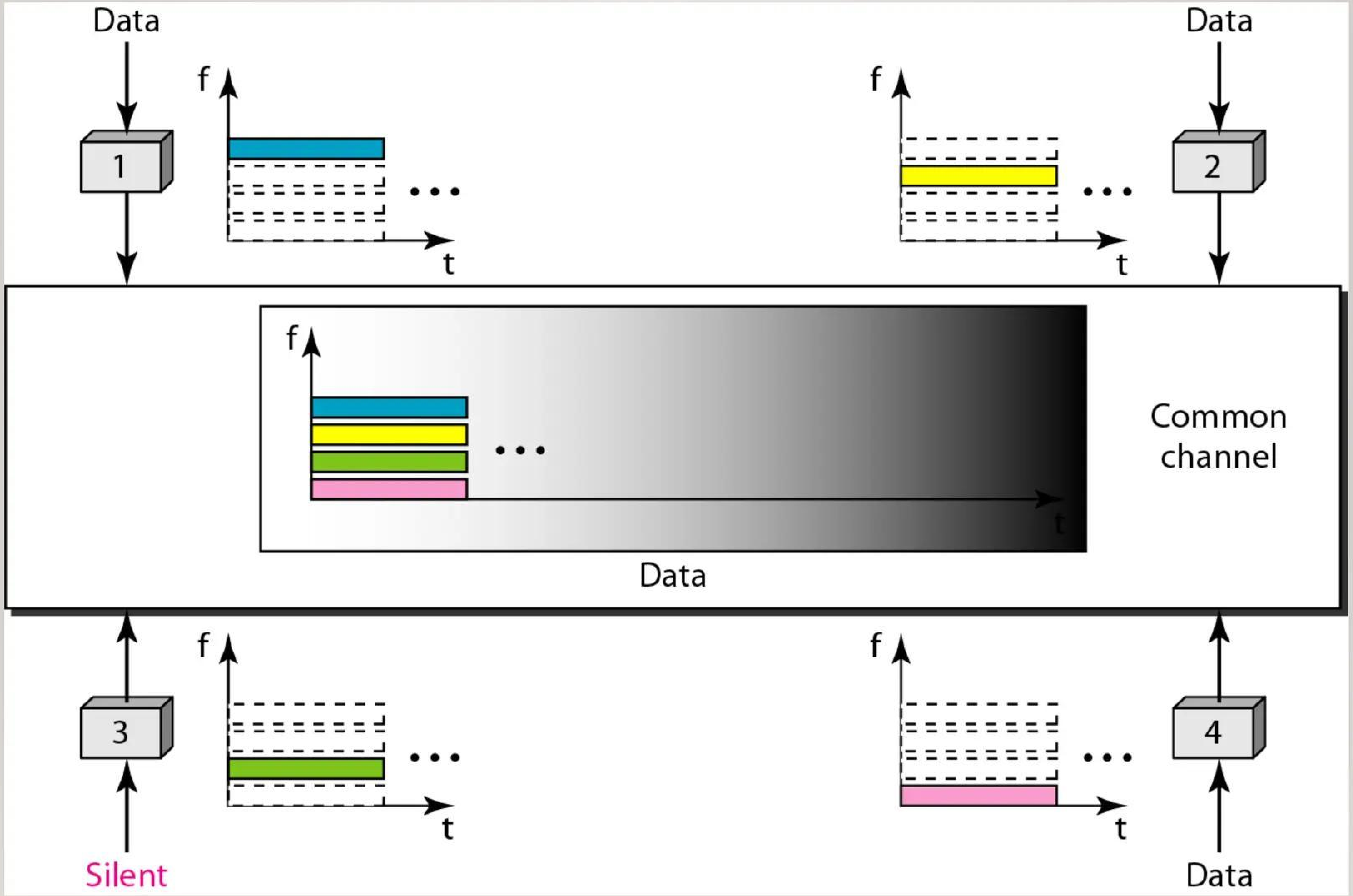


Figure 19 Frequency-division multiple access (FDMA)

Frequency-division multiple access (FDMA)

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

Time-division multiple access (TDMA)

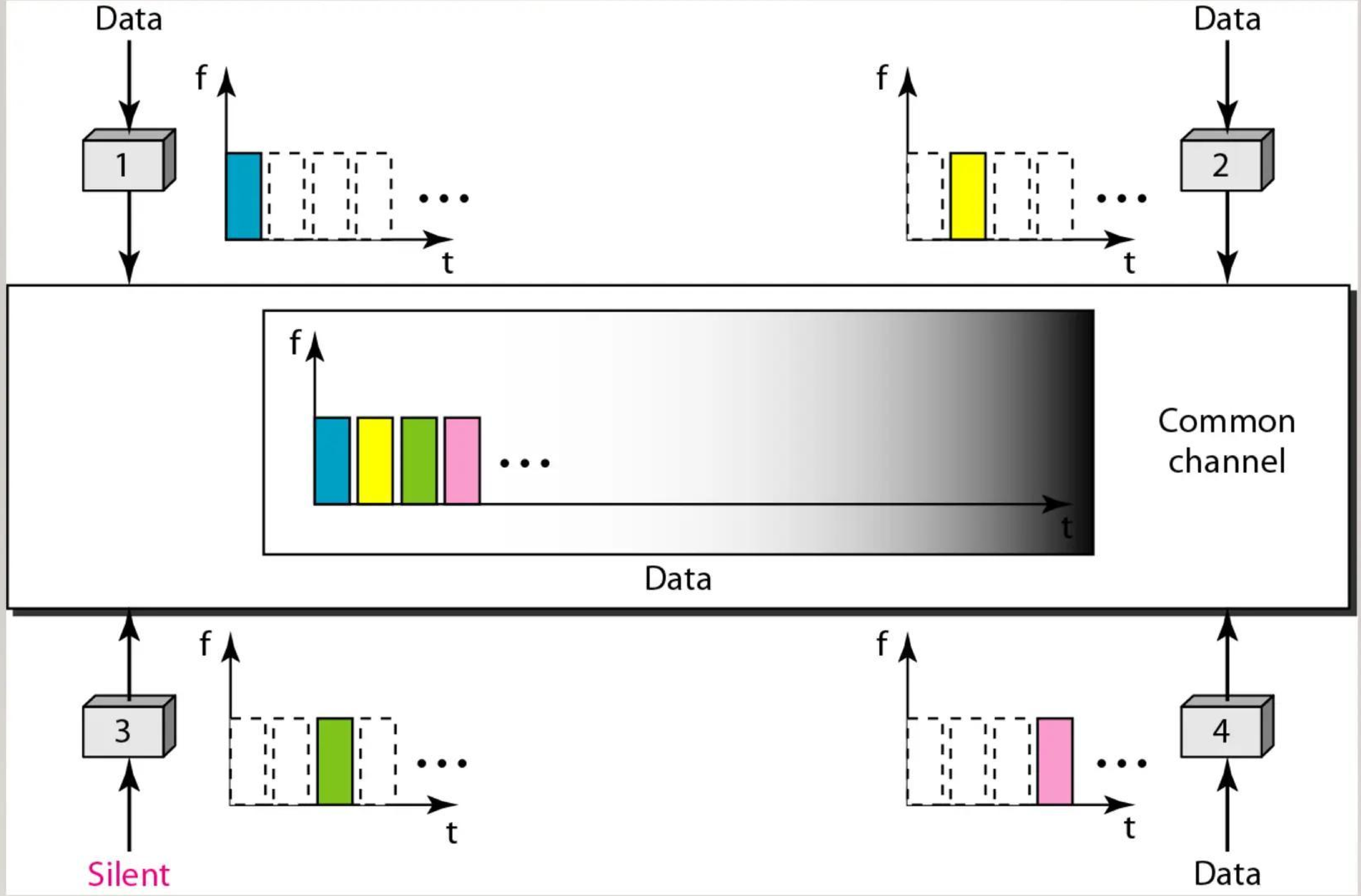


Figure 20 Time-division multiple access (TDMA)

Time-division multiple access (TDMA)

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

Code-division multiple access (CDMA)

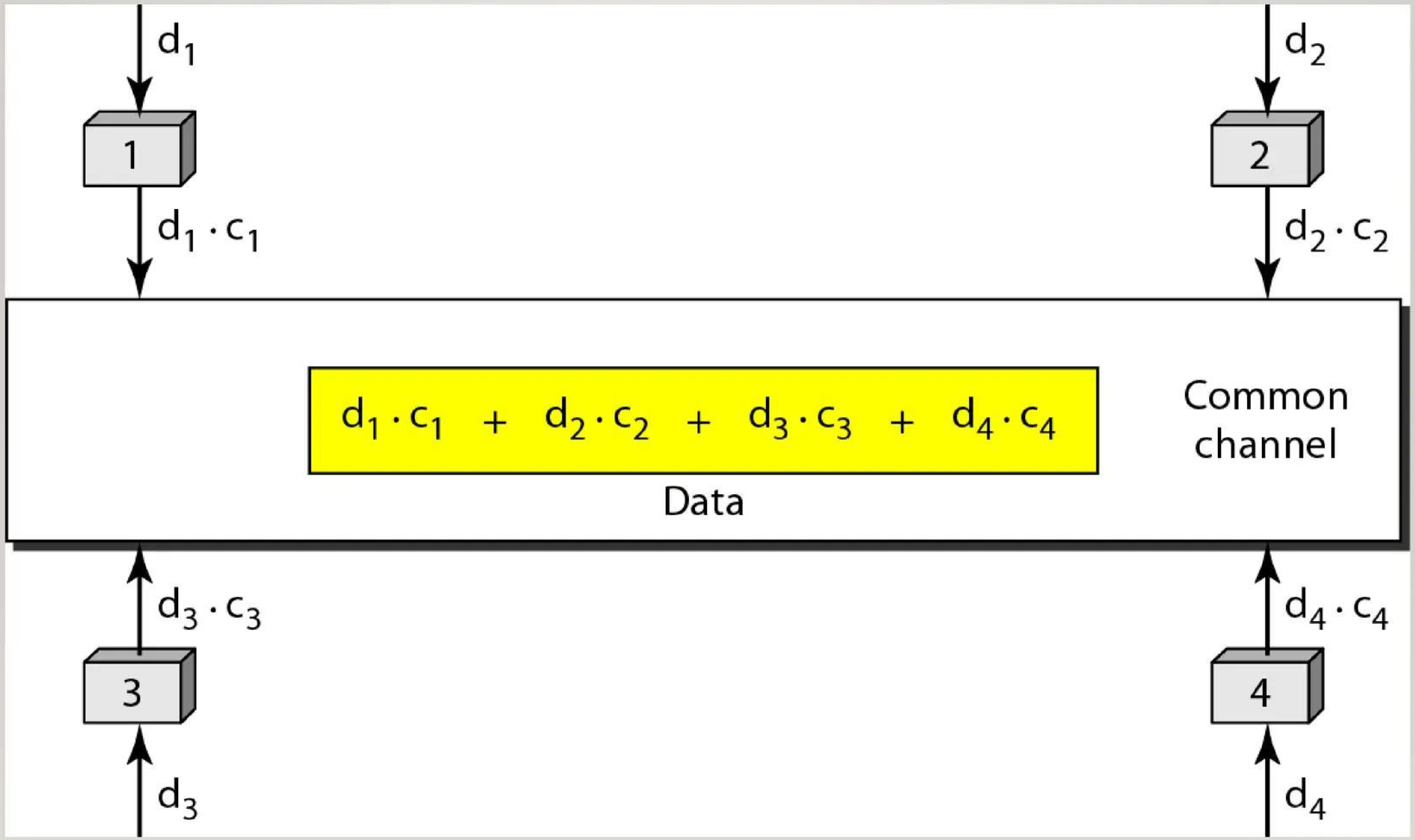


Figure 21 Simple idea of communication with code

Chip sequences

The following figure shows the Chip sequences:

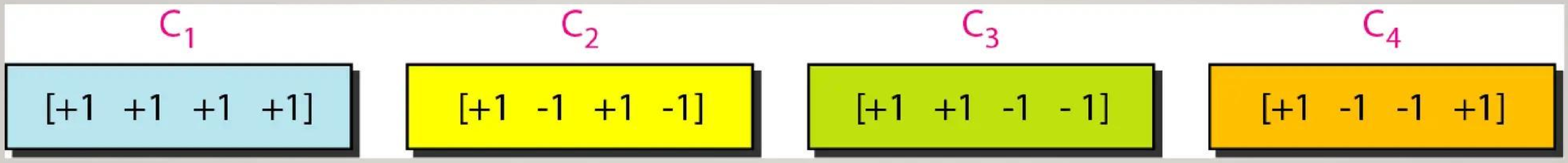


Figure. 22 Chip sequences

Data representation in CDMA

The following figure shows the data representation in CDMA:

Figure 23 Data representation in CDMA

Sharing channel in CDMA

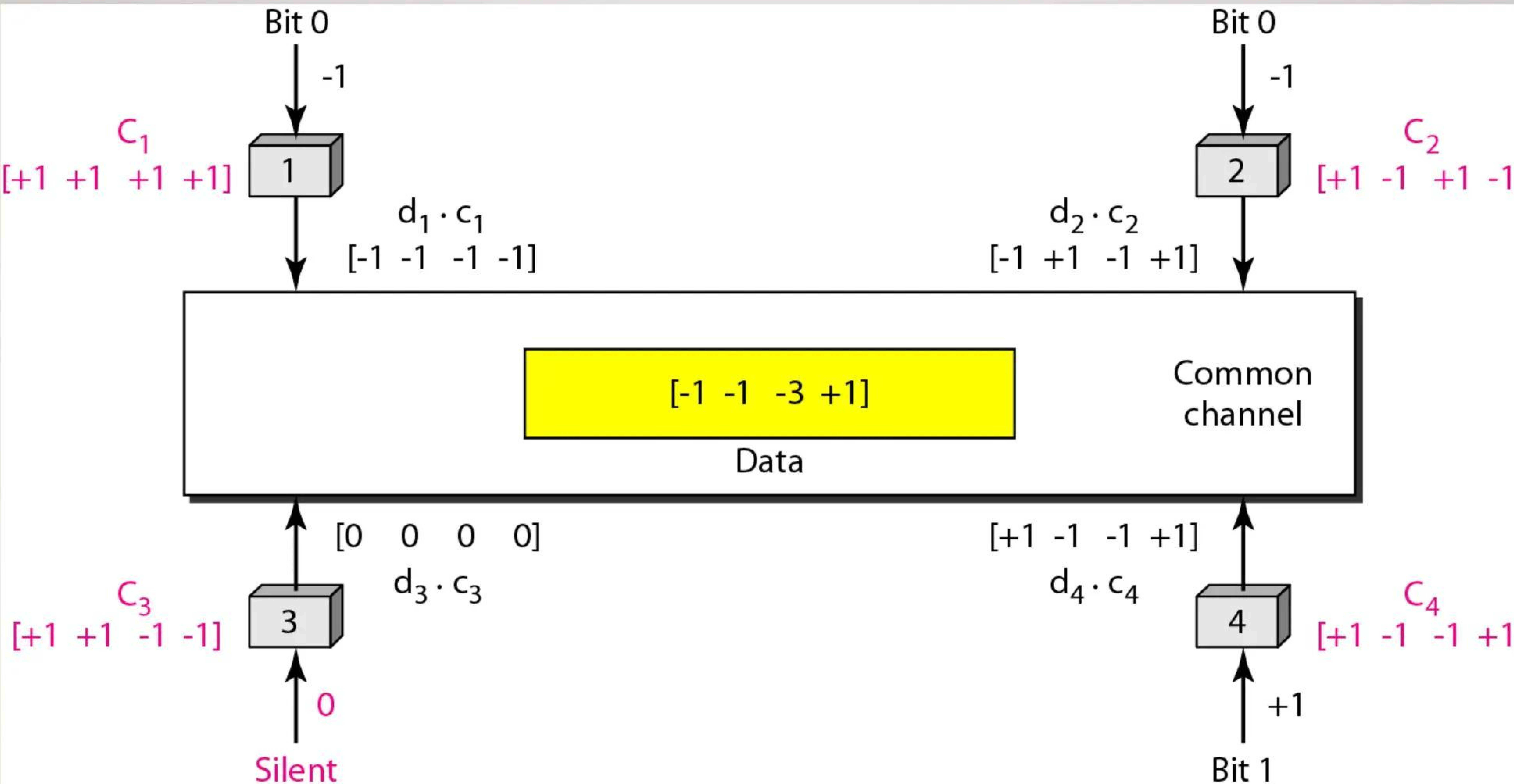


Figure24 Sharing channel in CDMA

Digital signal created by four stations in CDMA

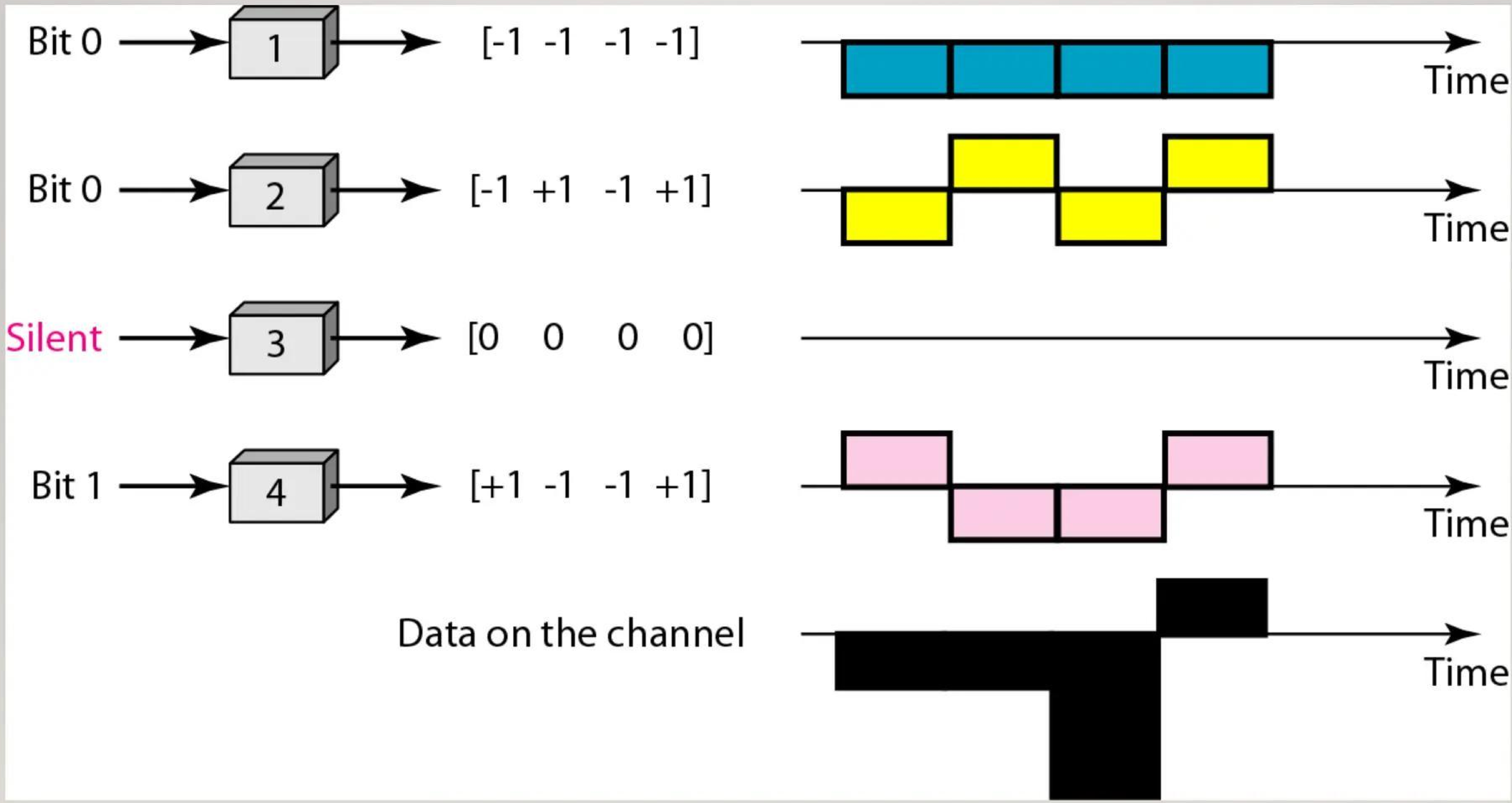


Figure 25 Digital signal created by four stations in CDMA

Decoding of the composite signal for one in CDMA

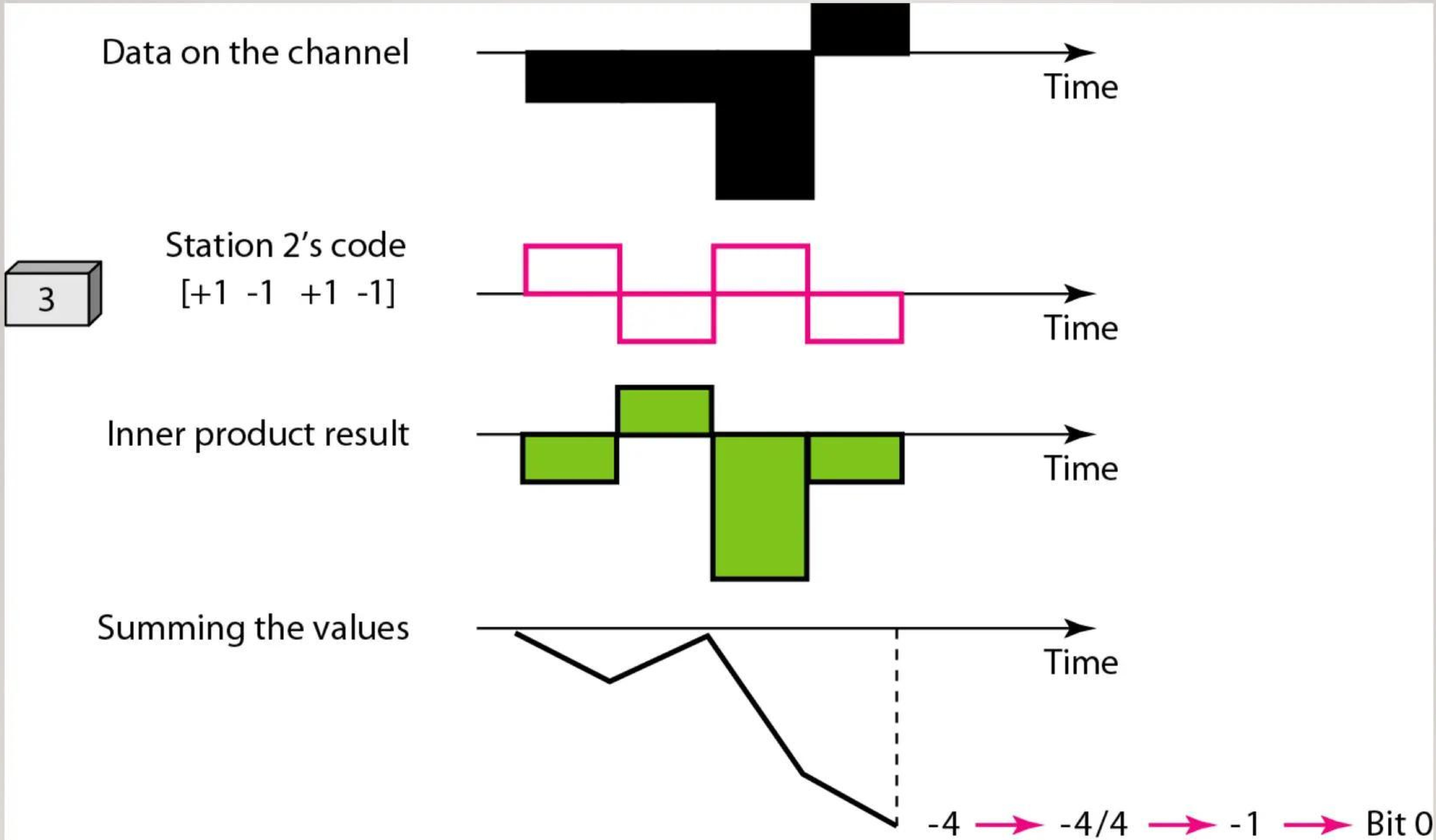


Figure 26 Decoding of the composite signal for one in CDMA

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happy

LEARNING